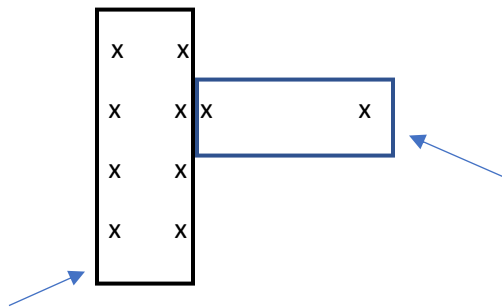


Worksheet 20 – Understanding Connectivity

(a) Is Connectivity an internal or external validation measure? What does connectivity measure?

(b) The textbook teaches you how to use the clValid package to produce the connectivity score but it does not explain how that value is obtained. The documentation for the clValid (a copy is found in Brightspace) provides information on how this score is calculated. The diagram below shows how 10 points are clustered, drawn in 2-D space with x and y being on the same scale (in other words, it means you can tell visually which distances are shorter and which are further). The arrows point to the 2 clusters. Is it possible to calculate the Connectivity score (using number of nearest neighbours = 2) without the actual values of the observations? If yes, calculate the Connectivity score.



(c) Now that you understand how Connectivity is calculated, can you tell if a low score or a high score is ideal? What is the lowest possible score and what is the highest possible connectivity score?

(d) Write an R program to validate that the calculated connectivity score in part (b) is correct. Write your code below.